

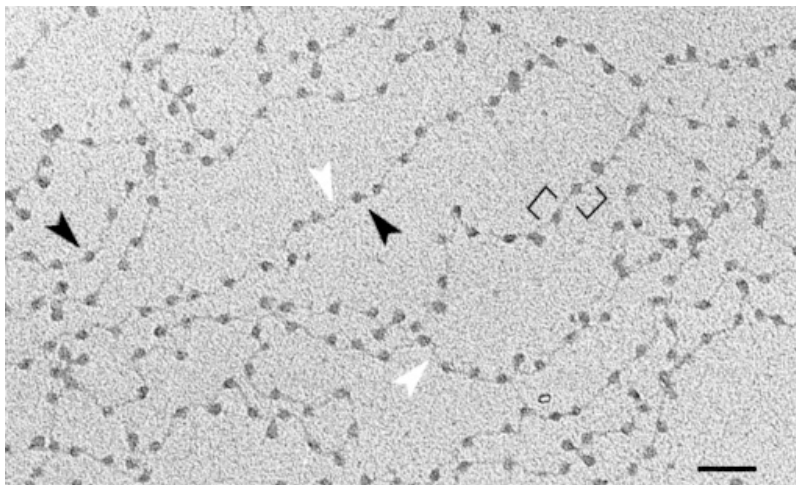
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**Question #:** 1

A researcher is studying the structure of chromatin using electron microscopy. An example of one of these images is shown which illustrates how DNA wraps around a group of proteins to form a nucleosome unit. The amino acid composition of these proteins helps in their physiological function. Which of the following amino acid residues most likely facilitates this interaction?

- A. Branched chain amino acids like leucine
- B. Negatively charged amino acids like aspartate
- C. Positively charged amino acids like lysine
- D. Thiol group containing amino acids like cysteine
- E. Hydroxyl group containing amino acids like serine
- F. Aromatic amino acids like phenylalanine

**Attachment:**



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**Question #:** 2

An investigator is analyzing the changes in a ciprofloxacin-resistant strain of *Pseudomonas aeruginosa*. The DNA gyrase of the bacteria is purified. It is shown to have a variant amino acid sequence when compared to the enzyme in the ciprofloxacin-sensitive strain. The drug no longer interacts with the variant DNA gyrase. Which of the following is the most likely function of the enzyme in the ciprofloxacin-sensitive strain?

- A. Cleaves single-stranded DNA
- B. Cleaves double-stranded DNA

- C. Facilitates formation of heterochromatin
- D. Catalyzes the formation of the peptide bond
- E. Causes termination of RNA synthesis

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**Question #: 3**

An investigator isolates a bacterial strain that is found to have reduced rates of mRNA synthesis. Which of the following proteins is most likely to exhibit reduced function in this strain?

- A. Dna A protein
- B. Sigma factor
- C. Single strand binding proteins (SSB)
- D. Initiation factor-1 (IF-1)
- E. Peptidyl transferase
- F. Elongation factor-G

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**Question #: 4**

A pharmaceutical company develops a drug that inhibits the enzyme which catalyzes the formation of a phosphodiester bond between a 3'-OH group of a nucleotide and a nucleoside monophosphate of a polynucleotide with ATP hydrolysis. Which of the following enzymes is most likely inhibited by this drug?

- A. DNA polymerase I
- B. DNA ligase
- C. Telomerase
- D. RNA polymerase
- E. Helicase
- F. Peptidyl transferase
- G. Aminoacyl tRNA synthetase

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**Question #: 5**

An enzyme with phenylalanine at amino acid position 206 is isolated. A variant enzyme is identified in the population in which a change in amino acid sequence at this position is observed, but there is no change in the charge and electrophoretic properties of the enzyme. Which of the following amino acids is most likely replaced in the variant enzyme?

- A. Lysine
- B. Glutamate
- C. Aspartate
- D. Tryptophan
- E. Arginine
- F. Histidine

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**Question #: 6**

A 15-day-old boy is brought to the emergency room by his parents because of a 30-minute history of seizures. Physical examination shows a lethargic infant. Laboratory studies show the presence of branched chain keto acids in urine. A special formula with restricted amounts of branched chain amino acids is prescribed. Dietary restriction of which of the following amino acids is most likely advised?

- A. Serine and glycine
- B. Glutamate and aspartate
- C. Leucine and valine
- D. Methionine and cysteine
- E. Lysine and leucine
- F. Cysteine and phenylalanine
- G. Arginine and histidine

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**Question #: 7**

A 7-year-old girl is diagnosed with type I diabetes mellitus. Genetic testing shows a point mutation, leading to the substitution of a highly conserved cysteine residue at amino acid position 7 in the B-chain of insulin. Which of the following is most likely affected as a result of this mutation?

- A. Formation of a disulfide bridge critical for folding of proinsulin
- B. Formation of hydrogen bonds between the A chain and B chain
- C. Formation of preproinsulin by the ribosome
- D. Removal of the amino-terminal signal sequence of preproinsulin
- E. Delivery of proinsulin to the endoplasmic reticulum
- F. Splicing of the insulin m-RNA

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**Question #: 8**

An investigator is analyzing tissue specific processing of the alpha-tropomyosin pre-mRNA which results in different mRNAs containing different combinations of introns and exons. It is observed that the alpha-tropomyosin protein in striated muscle is different compared to the alpha-tropomyosin protein in smooth muscle. Which of the following processes is most likely responsible for this difference?

- A. RNA editing
- B. Post-transcriptional modification
- C. Post-translational modification
- D. Proteolytic cleavage
- E. Epigenetic modification

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**Question #: 9**

A 2-year-old girl is brought to the physician by her parents because of a 1-month history of episodes of lethargy that occur approximately 4 hours following consumption of fruits. She is diagnosed with an inherited disorder of fructose metabolism. Sequencing of the mRNA coding for an enzyme required for fructose metabolism shows a single point mutation in a codon (AGU? AGA). Which of the following amino acids is most likely present at this position in the altered protein in the child?

- A. A negatively charged amino acid
- B. A sulfur containing amino acid
- C. A hydroxyl group containing amino acid
- D. A positively charged amino acid
- E. An aromatic amino acid
- F. A branched-chain amino acid
- G. A hydrophobic amino acid
- H. An acidic amino acid

**Attachment:**

|   | U   | C   | A    | G    |   |
|---|-----|-----|------|------|---|
| U | Phe | Ser | Tyr  | Cys  | U |
|   | Phe | Ser | Tyr  | Cys  | C |
|   | Leu | Ser | STOP | STOP | A |
|   | Leu | Ser | STOP | Trp  | G |
| C | Leu | Pro | His  | Arg  | U |
|   | Leu | Pro | His  | Arg  | C |
|   | Leu | Pro | Gln  | Arg  | A |
|   | Leu | Pro | Gln  | Arg  | G |
| A | Ile | Thr | Asn  | Ser  | U |
|   | Ile | Thr | Asn  | Ser  | C |
|   | Ile | Thr | Lys  | Arg  | A |
|   | Met | Thr | Lys  | Arg  | G |
| G | Val | Ala | Asp  | Gly  | U |
|   | Val | Ala | Asp  | Gly  | C |
|   | Val | Ala | Glu  | Gly  | A |
|   | Val | Ala | Glu  | Gly  | G |

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**Question #:** 10

A 56-year-old man comes to the physician because of a 2-day history of fever and a productive cough. His temperature is 39.8°C (100°F) and respirations are increased. An x-ray of the chest shows bilateral consolidation confirming pneumonia. Culture and sensitivity studies show the growth of erythromycin sensitive bacteria. Erythromycin is administered intravenously. Which of the following processes is most likely inhibited by this antimicrobial agent?

- A. Peptide bond formation during translation
- B. Initiation of translation
- C. Initiation of transcription
- D. DNA replication and cell division
- E. The translocation step of translation
- F. Unwinding of double-stranded DNA

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**Question #:** 11

An 8-month-old boy is brought to the physician by his mother because he is unable to sit without support. Physical examination shows a loss of motor skills and a delay in the developmental milestones. Ophthalmoscopy shows a 'cherry-red' spot on the retina. Laboratory studies show no detectable beta-hexosaminidase A activity and a diagnosis of Tay Sachs disease is made. Targeted gene mutation analysis shows a frameshift mutation in the gene encoding beta-hexosaminidase A. Which of the following mutations is most likely in this patient?

- A. Insertion of three nucleotides in an exon
- B. Deletion of three nucleotides in an exon
- C. Substitution of C for an A in an exon
- D. Insertion of four nucleotides in an exon
- E. Insertion of one nucleotide in an intron

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**Question #:** 12

A researcher studying carbohydrate metabolism isolates a bacterial strain which has a mutation in a gene encoding an enzyme required for fructose metabolism. There is no detectable enzyme activity in cell lysate from this bacterial strain. DNA sequencing of the gene encoding this protein shows that a UCA codon was mutated to a UAA codon. Which of the following best explains why this mutation led to a loss of enzyme activity in this bacterial strain?

- A. Ribosome is no longer able to initiate protein synthesis
- B. A frame shift was caused by the mutation
- C. An incorrect amino acid is placed in the enzyme active site
- D. A shorter than normal protein is synthesized
- E. There is defective splicing of the introns
- F. The mRNA is shorter than normal

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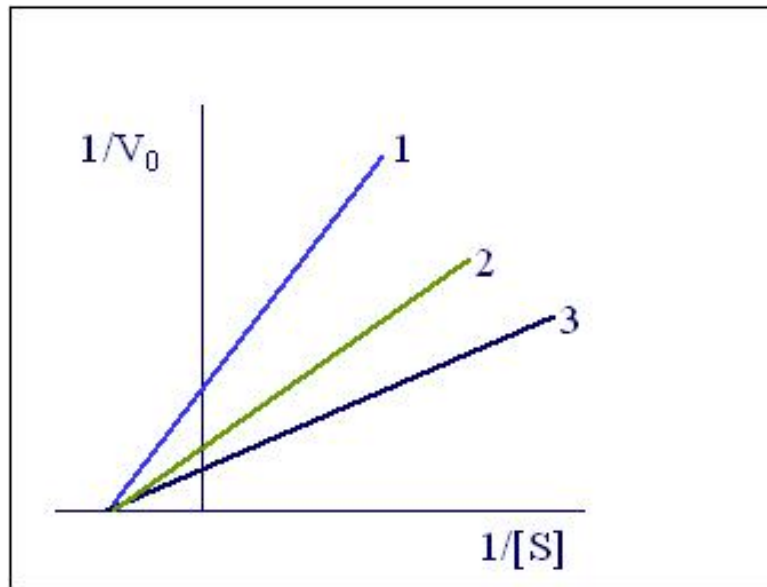
**Question #:** 13

An investigator is studying the kinetic properties of an enzyme in the presence of a newly discovered drug that acts as an enzyme inhibitor. The Lineweaver-Burk plots for the uninhibited reaction (no inhibitor added) and differing concentrations of the specific inhibitor is shown. Which of the following statements best describes the data shown in line 1?

- A. No inhibitor added
- B. Highest concentration of a competitive inhibitor

- C. Highest concentration of a noncompetitive inhibitor
- D. Lowest concentration of a competitive inhibitor
- E. Lowest concentration of a noncompetitive inhibitor

Attachment:



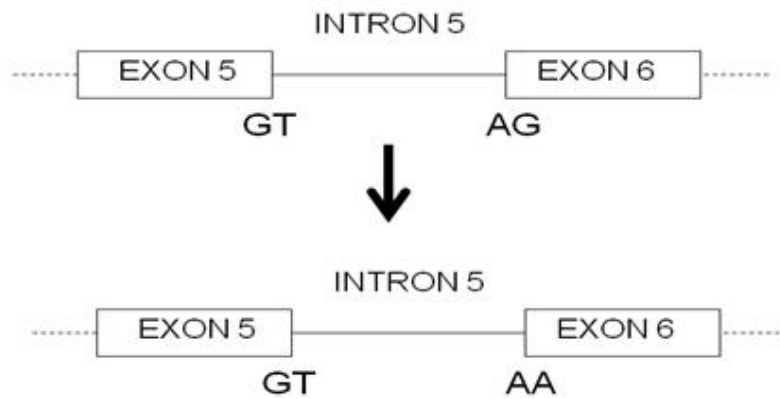
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Question #: 14

A 10-year-old boy is brought to the physician by his mother because of a 6-month history of progressive weakness of his arms and legs. Physical examination shows reduced muscle strength in the arms and hips and atrophy of the deltoid and gluteus muscles. DNA is prepared from a muscle biopsy obtained from the patient and regions of dysferlin are amplified by PCR. A mutation in dysferlin is identified and a diagnosis of Limb-Girdle Muscular Dystrophy 2B is made. The dysferlin gene is found to have an AG → AA alteration at position 1686 at the splice acceptor site (3' splice site) on intron 5 as shown in the figure. Which of the following is the most likely effect of this mutation?

- A. Exon 5 is absent in the mature mRNA
- B. The mature mRNA contains extra exons
- C. There is no change in the mature mRNA
- D. The mRNA has a premature stop codon in exon 5
- E. The protein has fewer amino acids

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**Question #: 15**

A bacterial strain is being studied that is found to have reduced rates of DNA replication resulting in reduced rates of cell division. Activity of which of the following proteins is most likely impaired in this strain?

- A. Sigma factor
- B. Single strand binding proteins (SSB)
- C. U1-snRNP
- D. Initiation factor-1 (IF-1)
- E. Peptidyl transferase

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**Question #: 16**

A researcher is analyzing the replicative DNA polymerase isolated from a novel bacterium. Studies show that the DNA polymerase from the organism lacks 3' to 5' exonuclease activity. Which of the following is most likely observed in this bacteria?

- A. Decreased mRNA synthesis
- B. Increased mutation rate
- C. Ability to replicate DNA without a primer
- D. Inability to replicate DNA
- E. Inability to unwind the double helical DNA

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**Question #: 17**

A 42-year-old woman comes to the physician because of a 4-day history of worsening cough and fever. Physical examination shows respiratory distress. Auscultation over the lungs shows diffuse crepitations and a diagnosis of lower respiratory tract infection is made. She is prescribed erythromycin for the infection. Which of the following processes is most likely inhibited by this drug?

- A. Binding of aminoacyl tRNA to A site of ribosome
- B. Unwinding of supercoiled DNA
- C. Initiation of translation
- D. Translocation of peptidyl tRNA
- E. Peptidyl transferase activity of ribosome
- F. Transcription of mRNA

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**Question #:** 18

During telophase at the end of mitosis, microtubules pull sister chromatids to opposite ends of the cell poles. Which best describes the portion of the chromosome that attaches to the microtubules?

- A. Euchromatin
- B. Heterochromatin
- C. Telomeric
- D. The *Alu* sequence
- E. Pseudogene

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**Question #:** 19

A scientist studying actively proliferating cultured cells uses a toxin to arrest the cells to block their exit from G<sub>2</sub>. Which of the following would best characterize these cells?

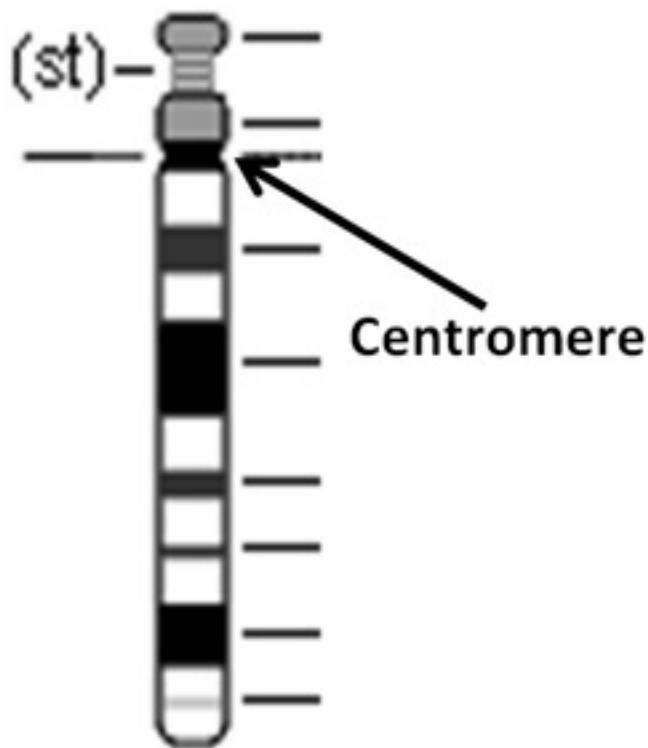
- A. Chromosomes are aligned at metaphase plate
  - B. Nuclear membrane has dissolved
  - C. The DNA is duplicated
  - D. Crossing over is occurring
  - E. Cell is preparing to duplicate DNA
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**Question #:** 20

A genetic test shows that the Down syndrome in a 4-year-old boy was caused by chromosome 21 undergoing a fusion to chromosome 14. Both of these chromosomes have a similar morphological structure as shown in the figure. Which of the following best describes the DNA that is found on the p arm of this type of chromosome?

- A. Methylated cytosine
- B. Repetitive ribosomal RNA genes
- C. Intermingled RNA/DNA
- D. Continuous centromeric structure
- E. Continuous telomeric structure

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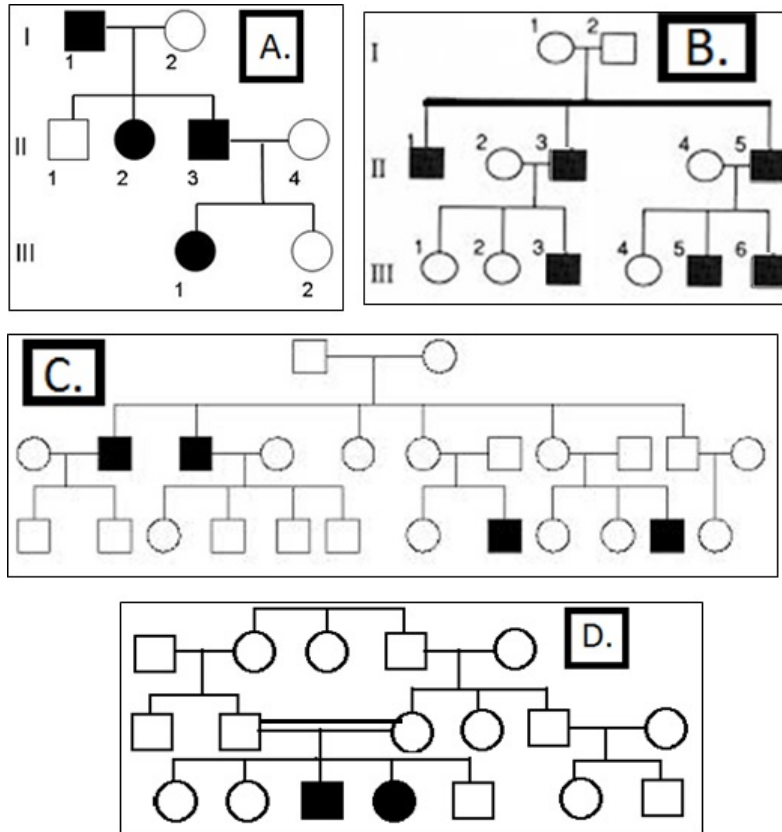
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**Question #:** 21

An investigator studying Mendelian traits constructs several pedigrees to describe classic patterns of inheritance. Which of the following genograms would most likely describe an autosomal recessive disorder?

- A. The pedigree in panel A
- B. The pedigree in panel B
- C. The pedigree in panel C
- D. The pedigree in panel D
- E. None of the pedigrees shown are suggestive of autosomal recessive

Attachment:



175209 ESoft Q fig ID.jpg

Question #: 22

A 22-year-old woman comes to the physician because she is concerned about a genetic condition that has affected some members of her family. Some of her relations have red-green color blindness and she wants to know if her children will be at risk of inheriting this trait. Which of the following pedigrees would most likely represent the family of this woman?

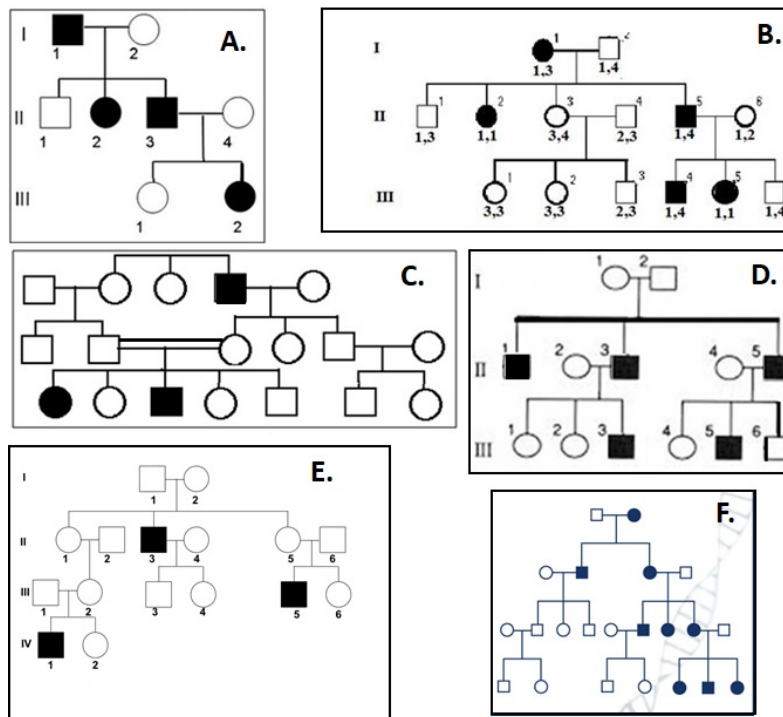
- A. Pedigree A
- B. Pedigree B
- C. Pedigree C
- D. Pedigree D
- E. Pedigree E

F. Pedigree F

G. All of these pedigrees are equally likely to represent this family

H. None of these pedigrees may represent this family

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### Question #: 23

A researcher finds that a protein truncation mutation in the lysine methyl transferase gene, *SETD6*, is responsible for a rare form of inherited colon cancer. The truncated *SETD6* protein is expressed in the same subcellular area, and it binds to its substrates as normal. However, the mutated protein blocks the ability of wild-type *SETD6* (expressed from the other allele) to function properly. This results in abnormal signaling and tumor development. Which of the following genetic terms best describes the mechanism of this inherited form of colon cancer?

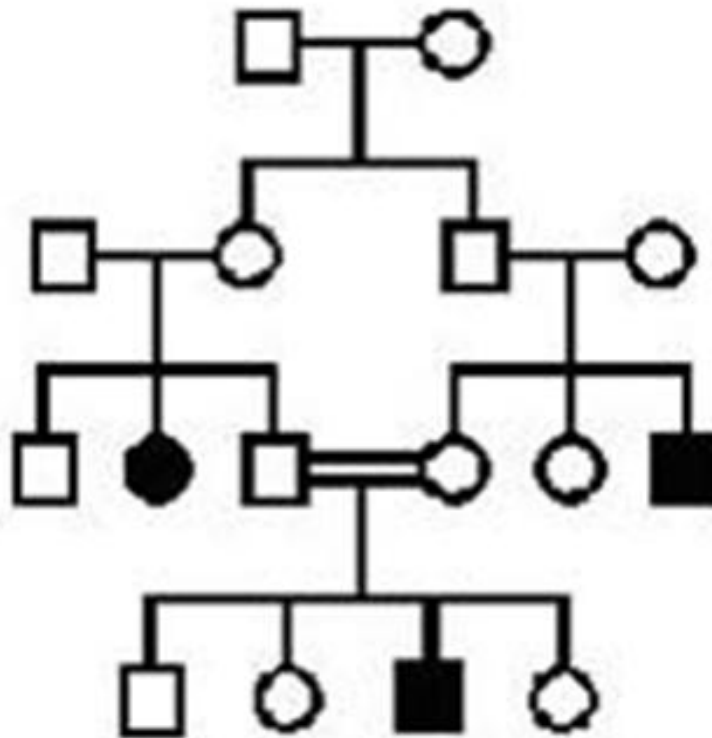
- A. Autosomal recessive
- B. Haploinsufficiency
- C. Dominant negative
- D. Anticipation
- E. X-linkage
- F. Imprinting

Question #: 24

A 42-year-old man (III-3) whose third child has phenylketonuria comes to a physician to learn about recurrent risk to his grandchildren as his first-born son now has a pregnant girlfriend. A pedigree of the family is shown (note that the pregnant girlfriend is not shown on this genogram). Which of the following would be the best estimate of the probability that IV-1 is a carrier of the mutation that leads to the inborn error of metabolism that was previously seen in this family?

- A.  $2/3$
- B.  $1/2$
- C. Virtually nil
- D. 100%
- E.  $1/4$

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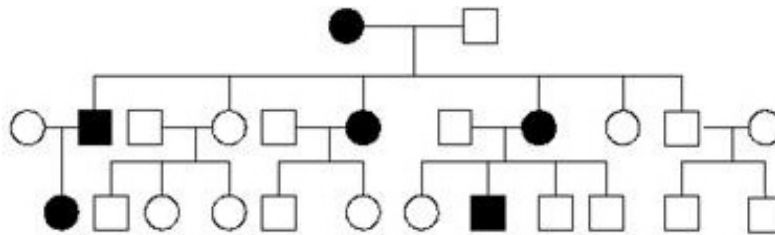
Question #: 25

A 23-year-old man comes to the physician for a follow up consultation about a previously diagnosed genetic disorder. He has arachnodactyly, pectus excavatum, and he is taller than his siblings and cousins. A pedigree of this patient's family is shown. At this visit, the physician

reviews the results from the latest echocardiogram because the patient is at risk for an aneurysm resulting from a dilated ascending aorta. Which of the following genes most likely has a pathogenic variant to cause the disorder in this patient?

- A. The *COL1A1* gene which encodes a pro-collagen chain
- B. The *FGF3R* gene which encodes fibroblast growth factor receptor 3
- C. The *LDLR* gene which encodes the LDL receptor
- D. The *FBN1* gene which encodes fibrillin
- E. The *DMD* gene which encodes dystrophin
- F. The *HTT* gene which encodes huntingtin
- G. The *DMPK* gene which encodes dystrophin myotonic protein kinase
- H. The *HEXA* gene which encodes hexosaminidase A

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**Question #:** 26

A 4-year-old boy is brought to the physician after worsening complaints of dizziness and headache. Laboratory tests show elevated insulin (hyperinsulinemia) and hypoglycemia. A genetic test shows a mutation in the gene encoding glucokinase. The effect of this mutation is predicted to mutationally alter the allosteric activator site of the enzyme. In which region of the gene would the disease-causing mutation most likely be found?

- A. Enhancer
- B. Exon
- C. Intron
- D. Promoter
- E. Shine Dalgarno

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**Question #:** 27

A 4-year-old girl is brought to the physician for follow-up examination by her parents because she shows phenotypic characteristics that are similar, but not identical, to Down Syndrome. Standard karyotype shows that in addition to trisomy 21, she has a large translocation where approximately 50% of the q arm of chromosome 1 is attached to another chromosome. Which of the following genetic terms would best describe the appearance of the aberrant chromosome 1 following this translocation event?

- A. Acrocentric
- B. Telocentric
- C. Barr body
- D. Submetacentric
- E. Metacentric

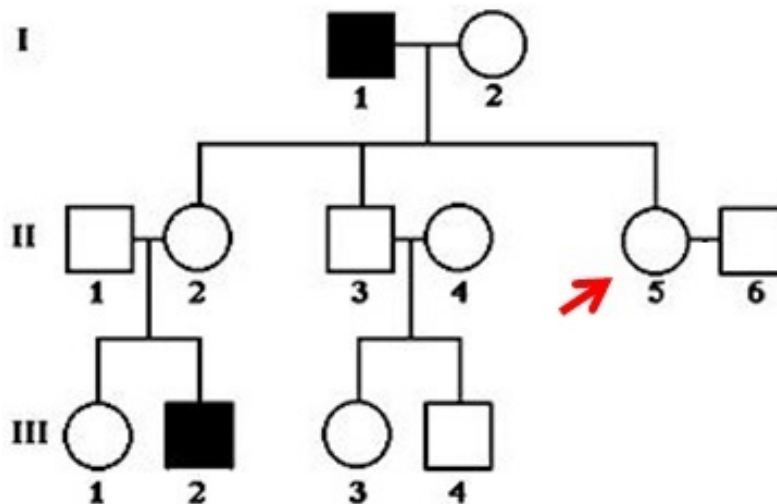
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**Question #:** 28

A 26-year-old woman (the consultand; II-5) comes to the physician because there is a history of a bleeding disorder due to deficiency of factor VIII in her family. She is now thinking about having a family of her own. A 3-generation pedigree of the immediate family is shown. Which of the following best explains why are there no affected family members in the second generation?

- A. Not enough female offspring in generation II (the sample size is too small)
- B. Normal X-linked inheritance
- C. Effect of heteroplasmy
- D. Phenotype exhibits variable expressivity
- E. The disorder exhibits incomplete penetrance
- F. The disorder exhibits allelic heterogeneity

**Attachment:**



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**Question #:** 29

A married couple comes to the physician to learn about the recurrence risk of cystic fibrosis (CF) to their future children. They are both healthy and currently have 1 child with CF. Analysis of the *CFTR* alleles of the father shows that he is heterozygous for the Phe508del variant (  $\Delta$ F508). Analysis of the mother shows that she is heterozygous for a missense variant leading to an Arg117His substitution. The child inherited the disease associated alleles from both parents. Which of the following genetic terms best describes this example?

- A. Incomplete penetrance
- B. Allelic heterogeneity
- C. Variable expression
- D. Heteroplasmy
- E. Locus heterogeneity
- F. Uniparental disomy

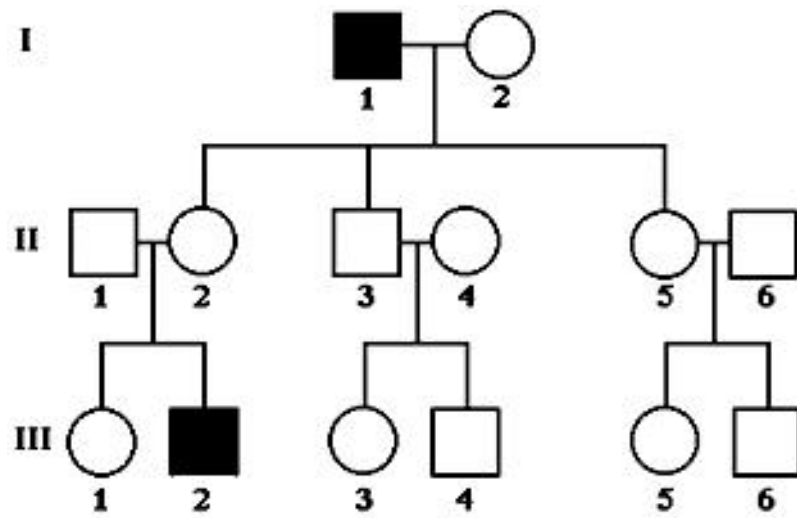
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**Question #:** 30

A 35-year-old woman comes to the physician for genetic counseling. She is a member of a family that has some individuals affected with Becker muscular dystrophy. Family history is recorded, and a pedigree of the family is shown. Which of the following individuals are most likely the obligate carriers of the pathogenic variant in this family?

- A. I-2, II-2
- B. II-1, III-1
- C. II-2, II-5
- D. III-5, II-4
- E. III-1, III-5
- F. II-2, II-3, II-5

**Attachment:**



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